

Report of PhD activities in the academic year
2017 / 2018

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Abstract

This document describes the achievements, current state and future plans of the research project, moreover it contains the courses and conferences attended in the 2017 / 2018 academic year.

Contents

1	Current state of the research topic - Atmosphere modelling	4
1.1	Achievements	4
1.2	Outlook	4
2	Courses, summer schools and seminars	4
2.1	Courses	4
2.1.1	Atmosphere dynamics - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila	4
2.1.2	Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI	4
2.2	Conferences, seminars, summer schools	4
2.2.1	Intensive Program on Fluids and Waves, GSSI	4
2.2.2	Women in applied and computational Mathematics, GSSI	5
2.2.3	Evans function methods in the stability analysis of peri- odic wavetrains, Prof. R. Plaza	5
2.2.4	On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza	5
2.2.5	Secondment at the University of Sussex, Prof. O. Lakkis .	5
2.2.6	Waves in Flows Summer School in Prague	5

1 Current state of the research topic - Atmosphere modelling

1.1 Achievements

1.2 Outlook

2 Courses, summer schools and seminars

2.1 Courses

2.1.1 Atmosphere dynamics - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila

Physics of the atmosphere.

2.1.2 Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI

This course gave an overview on shock capturing schemes for the numerical solution of systems of conservation and balance laws. The emphasis was on the construction of high order finite volume and finite difference schemes, which are based on the three main ingredients: numerical flux, high order non oscillatory reconstruction, and high order integration in time. The first part of the course illustrated some models in which the nonlinearity produces shocks that propagate in time. Simple three point schemes, such as upwind, Lax-Friedrichs, and Lax-Wendroff was reviewed, and their behaviour on linear equations and systems analysed, before the illustration of more complex methods for the quasi-linear systems. The step by step construction of the schemes was illustrated in lab sessions, and the methods were applied to concrete examples of physical interest. The last part of the course was devoted to systems with source term.

2.2 Conferences, seminars, summer schools

2.2.1 Intensive Program on Fluids and Waves, GSSI

This training activity within the European Innovating Training Network Mod-CompShock surveyed recent results in the analysis of fluids and waves with emphasis on modelling and computational methods as well as its theoretical aspects.

- i) Week 1. Workshop/School in Numerics for Fluids and Waves
- ii) Week 2. Mini courses by J. Bedrossian. and D. Cordoba
- iii) Week 3. Mini courses by C. Bardos, Z. Grujic, J. Malek
- iv) Week 4. Workshop on Mathematical fine structures in fluid dynamics

2.2.2 Women in applied and computational Mathematics, GSSI

This three-day event was designed for researchers challenged by the application of various techniques in modern analysis and computational mathematics to intriguing problems arising in physics, material science and biology. The topic and the results of this present research project were represented - together with the work of many young female researchers - in the poster session.

2.2.3 Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza

This lecture series provided an introduction to recent techniques in the spectral stability analysis of spatially periodic travelling waves which involve the periodic Evans function introduced by Gardner.

2.2.4 On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza

2.2.5 Secondment at the University of Sussex, Prof. O. Lakkis

The aim of this 6-week secondment was to create both the theoretical and technical foundations for numerical simulations in order to visualise the results of the project. The main results and steps are the following:

- i) Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed
- ii) Writing scripts to login / connect to the HPC server at the Sussex University
- iii) Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs
- iv) Creating foundation for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

2.2.6 Waves in Flows Summer School in Prague

The aim of this summer school was to present a comprehensive series of lectures on various aspects of waves and their role in mathematical analysis and numerical simulation of different types of flows. The school was organised as a multidisciplinary event with topics ranging from theoretical and numerical mathematics up to physics and applications. The emphasis was on models related to topics originating in environmental, biomedical and industrial applications. Besides of the course lectures, the program included a one day workshop dedicated to short lectures given by attending scientists.

References

- [Adams and Fournier, 2003] Adams, R. A. and Fournier, J. J. (2003). *Sobolev spaces*, volume 140. Academic press.
- [Ali et al., 1998] Ali, I., Kalla, S., and Khajah, H. (1998). A partial differential equation related to a problem in atmospheric pollution. *Mathematical and computer modelling*, 28(12):1–6.
- [Anderson and Wendt, 1995] Anderson, J. D. and Wendt, J. (1995). *Computational fluid dynamics*, volume 206. Springer.
- [Andria et al., 2000] Andria, G., Lay-Ekuakille, A., and Notarnicola, M. (2000). Mathematical models for atmospheric and industrial pollutant prediction. In *XVI IMEKO World Congress, Vienna, Austria*.
- [Arya, 2001] Arya, P. S. (2001). *Introduction to micrometeorology*, volume 79. Elsevier.
- [Azérad and Guillén, 2001] Azérad, P. and Guillén, F. (2001). Mathematical justification of the hydrostatic approximation in the primitive equations of geophysical fluid dynamics. *SIAM Journal on Mathematical Analysis*, 33(4):847–859.
- [Besson and Laydi, 1992] Besson, O. and Laydi, M. (1992). Some estimates for the anisotropic navier-stokes equations and for the hydrostatic approximation. *ESAIM: Mathematical Modelling and Numerical Analysis*, 26(7):855–865.
- [de Swart, 2016] de Swart, H. E. (2016). The governing equations and the dominant balances of flow in the atmosphere and ocean. *USPC Summerschool, Utrecht*.
- [Goyal and Kumar, 2011] Goyal, P. and Kumar, A. (2011). Mathematical modeling of air pollutants: An application to indian urban city. In *Air Quality-Models and Applications*. InTech.
- [Hosseini, 2013] Hosseini, B. (2013). Dispersion of pollutants in the atmosphere: a numerical study. Master’s thesis, Simom Fraser University.
- [Kathirgamanathan et al., 2002] Kathirgamanathan, P., McKibbin, R., and McLachlan, R. (2002). Source term estimation of pollution from an instantaneous point source. *Research Letters in Information Mathematic Science*, Vol. 3, pp. 59-67.
- [Lions, 1996] Lions, P.-L. (1996). *Mathematical Topics in Fluid Mechanics: Volume 2: Compressible Models*, volume 2. Oxford University Press on Demand.
- [Marsaleix et al., 2006] Marsaleix, P., Auclair, F., and Estournel, C. (2006). Considerations on open boundary conditions for regional and coastal ocean models. *Journal of Atmospheric and Oceanic Technology*, 23(11):1604–1613.

- [Monin, 1959] Monin, A. (1959). On the boundary condition on the earth surface for diffusing pollution. In *Advances in Geophysics*, volume 6, pages 435–436. Elsevier.
- [Pedlosky, 2013] Pedlosky, J. (2013). *Geophysical fluid dynamics*. Springer Science & Business Media.
- [Prodanova et al., 2008] Prodanova, M., Perez, J. L., Syrakov, D., San Jose, R., Ganev, K., Miloshev, N., and Roglev, S. (2008). Application of mathematical models to simulate an extreme air pollution episode in the bulgarian city of stara zagora. *Applied Mathematical Modelling*, 32(8):1607–1619.
- [Simon, 1986] Simon, J. (1986). Compact sets in the space $l^p(0, t; b)$. *Annali di Matematica pura ed applicata*, 146(1):65–96.
- [Temam, 2001] Temam, R. (2001). *Navier-Stokes equations: theory and numerical analysis*, volume 343. American Mathematical Soc.
- [Temam and Ziane, 2005] Temam, R. and Ziane, M. (2005). Some mathematical problems in geophysical fluid dynamics. *Handbook of mathematical fluid dynamics*, 3:535–658.
- [Wang et al., 2014] Wang, Q., Zhou, W., Wang, D., and Dong, D. (2014). Ocean model open boundary conditions with volume, heat and salinity conservation constraints. *Advances in Atmospheric Sciences*, 31(1):188–196.
- [Yeh and Huang, 1975] Yeh, G.-T. and Huang, C.-H. (1975). Three-dimensional air pollutant modeling in the lower atmosphere. *Boundary-Layer Meteorology*, 9(4):381–390.
- [Zhuk et al., 2016] Zhuk, S., Tchrakian, T. T., Moore, S., Ordóñez-Hurtado, R., and Shorten, R. (2016). On source-term parameter estimation for linear advection-diffusion equations with uncertain coefficients. *SIAM Journal on Scientific Computing*, 38(4):A2334–A2356.